

Experiment 04:

Laboratory Exercise

1. identify available storage devices and partitions to avoid imaging the wrong device.

```
(student@kali)-[/home/student/Shivam]
PS> lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda          8:0    0  100G  0 disk
├─sda1       8:1    0   96G  0 part /
├─sda2       8:2    0    1K  0 part
└─sda5       8:5    0    4G  0 part [SWAP]
sr0         11:0    1 1024M  0 rom
```

Observation:

The **lsblk** command displayed all available storage devices, their partitions, and mount points. This helped identify the correct storage device before creating the forensic image.

2. Create a Directory for storing the evidence.

```
(student@kali)-[/home/student/Shivam]
PS> mkdir forensic_evidence

(student@kali)-[/home/student/Shivam]
PS> ls
cloudflare.com_ips.txt  forensic_evidence  new1.txt  new.txt  Radha.txt  Shivam.txt
```

Observation:

A new folder named **forensic_evidence** was created successfully. The **ls** command confirmed that the folder was present in the current directory and ready to store the image file.

3. Create a small directory to simulate evidence.

```
(student@kali)-[/home/student/Shivam]
PS> mkdir sample_data

(student@kali)-[/home/student/Shivam]
PS> ls
cloudflare.com_ips.txt  forensic_evidence  new1.txt  new.txt  Radha.txt  sample_data  Shivam.txt
```

Observation:

A folder named **sample_data** was created successfully. This folder was used to store sample files that acted as digital evidence for the experiment.

4. Create an archive as we are imaging a directory rather than a disk and create a forensic image.

```
(student@kali)-[/home/student/Shivam]
PS> echo "Digital Forensics Lab" > sample_data/file1.txt

(student@kali)-[/home/student/Shivam]
PS> echo "Ethical Hacking" > sample_data/file2.txt

(student@kali)-[/home/student/Shivam]
PS> echo "Forensic Duplicate" > sample_data/file3.txt
```

Observation:

Three sample text files were created inside the **sample_data** folder. The folder was archived into a TAR file, and a forensic image named **evidence.dd** was created successfully from the archive.

5. Verify Successful Image Creation.

```
(student@kali)-[/home/student/Shivam]
PS> ls sample_data
file1.txt  file2.txt  file3.txt

(student@kali)-[/home/student/Shivam]
PS> tar -cvf sample_data.tar sample_data
sample_data/
sample_data/file3.txt
sample_data/file1.txt
sample_data/file2.txt

(student@kali)-[/home/student/Shivam]
PS> dd if=sample_data.tar of=forensic_evidence/evidence.dd bs=4M status=progress
0+1 records in
0+1 records out
10240 bytes (10 kB, 10 KiB) copied, 0.000667019 s, 15.4 MB/s

(student@kali)-[/home/student/Shivam]
PS> ls forensic_evidence
evidence.dd

(student@kali)-[/home/student/Shivam]
PS> ls -l forensic_evidence
total 12
-rw-rw-r-- 1 student student 10240 Aug  7 11:52 evidence.dd
```

Observation:

The evidence.dd file was found inside the **forensic_evidence** folder after the imaging process. This confirmed that the forensic image was created successfully without any errors.

6. Calculate hash of the acquired image.

```
(student@kali)-[/home/student/Shivam]
PS> sha256sum forensic_evidence/evidence.dd
a081895f9cb9e70fae802c3e946b0720cb01237e299284e6b91fbaff08978179  forensic_evidence/evidence.dd
```

Observation:

The sha256sum command generated a unique **SHA-256** hash value for the image file. This hash can be used later to verify that the image has not been changed.

7. Verify image file information.

```
(student@kali)-[/home/student/Shivam]
PS> file forensic_evidence/evidence.dd
forensic_evidence/evidence.dd: POSIX tar archive (GNU)
```

Observation:

The file command showed that the created image was a valid **POSIX tar archive**. This confirmed that the image file format was correct.

8. Verify the image size.

```
(student@kali)-[/home/student/Shivam]
PS> ls -lh forensic_evidence/evidence.dd
-rw-rw-r-- 1 student student 10K Aug  7 11:52 forensic_evidence/evidence.dd
```

Observation:

The **ls -lh** command displayed the size and other details of the image file. The image size matched the expected output shown during the experiment.

9. TAR-based image, first create a mount directory and extract read-only copy for examination.

```
(student@kali)-[/home/student/Shivam]
PS> tar -xvf forensic_evidence/evidence.dd -C mount_evidence
sample_data/
sample_data/file3.txt
sample_data/file1.txt
sample_data/file2.txt
```

```
(student@kali)-[/home/student/Shivam]
PS> find mount_evidence
mount_evidence
mount_evidence/sample_data
mount_evidence/sample_data/file3.txt
mount_evidence/sample_data/file1.txt
mount_evidence/sample_data/file2.txt
```

Observation:

A folder named **mount_evidence** was created, and the image was extracted into it. All the sample files were successfully restored and were ready for examination.

10. View Contents of the Mounted Image.

```
(student@kali)-[/home/student/Shivam]
PS> cat mount_evidence/sample_data/file1.txt
Digital Forensics Lab
```

Observation:

The contents of the extracted file were displayed using the **cat** command. The text stored in the file matched the original data, showing that the image was created correctly.

11. Successful unmount the Image.

```
(student@kali)-[/home/student/Shivam]
PS> rm -r mount_evidence

(student@kali)-[/home/student/Shivam]
PS> ls
cloudfore.com_ips.txt  forensic_evidence  new1.txt  new.txt  Radha.txt  sample_data
```

Observation:

The **mount_evidence** directory was removed successfully after the examination was completed. This confirmed that the extracted files were cleaned up properly and the experiment ended successfully.